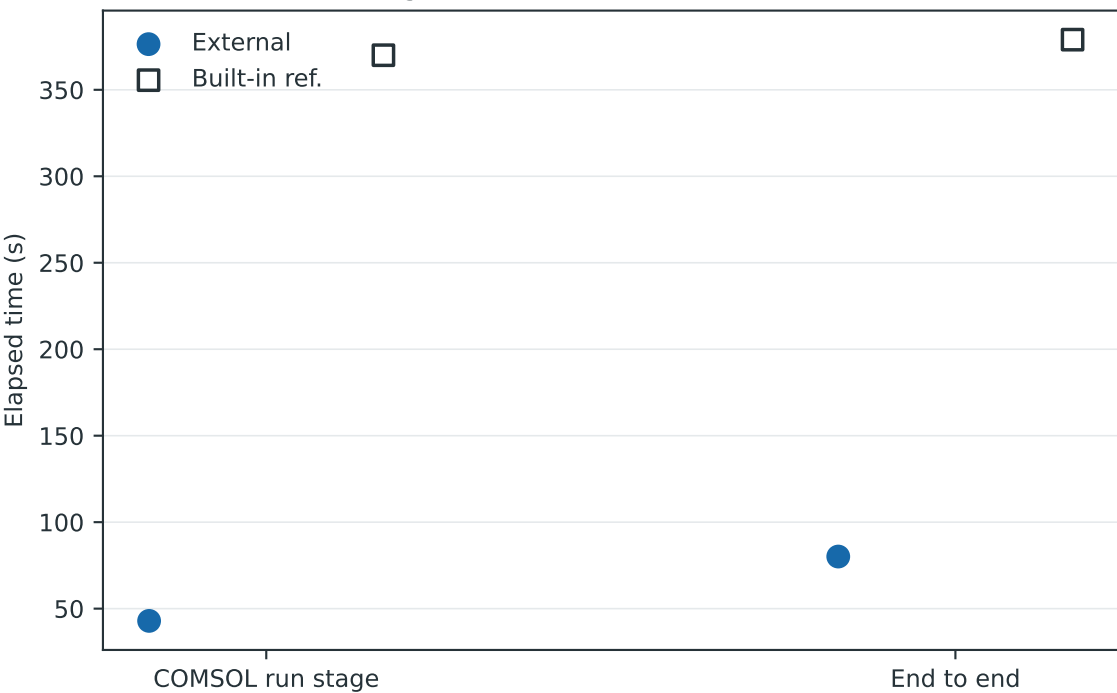


Runtime observations and external workflow breakdown
provisional: minimum n=1, required n=3; solver architectures, DOF, and continuation are unmatched

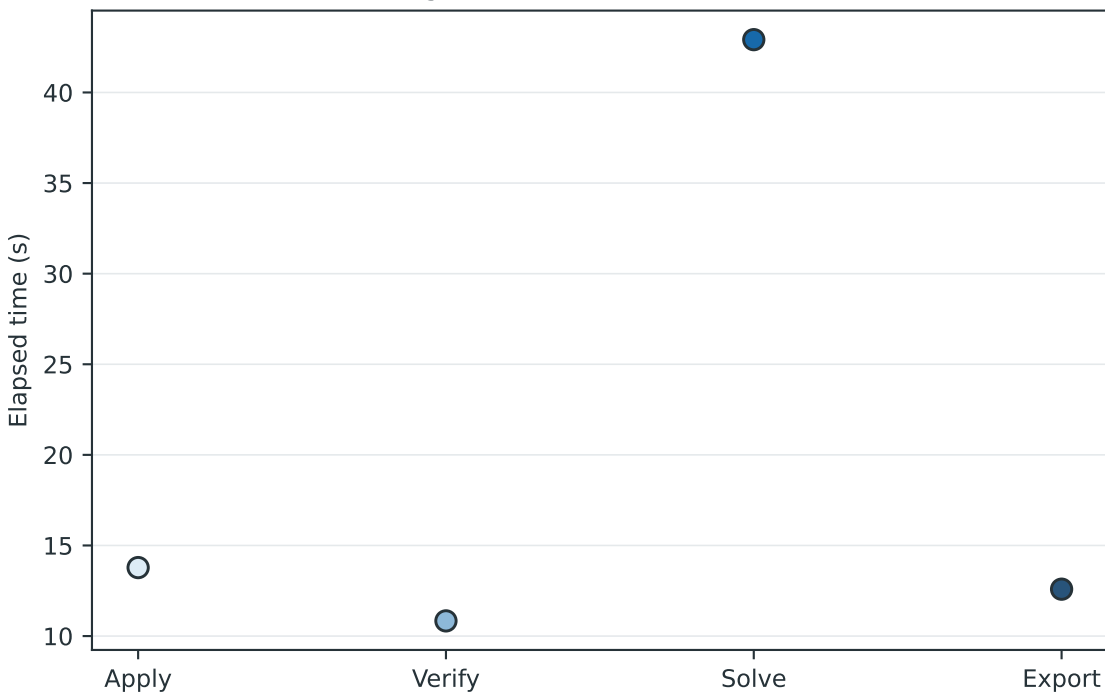
A

Path runtime: run-stage and end-to-end observations



B

External end-to-end stage breakdown



Source: runtime observations CSV. Markers are single observations (n=1); no distribution or uncertainty interval is shown. Historical 42.914 s is a COMSOL run-stage time, not isolated nonlinear-solver time; the CSV does not bind the 370 s entry to an identified raw log. Evidence status: provisional_2026-07-14.